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CYBER SECURITY LAB EXPERIMENT

Title: Privacy Audit of Popular Apps (Desktop **WhatsApp**) and Websites (**Facebook**) & Data Breach Case Study Analysis

Aim

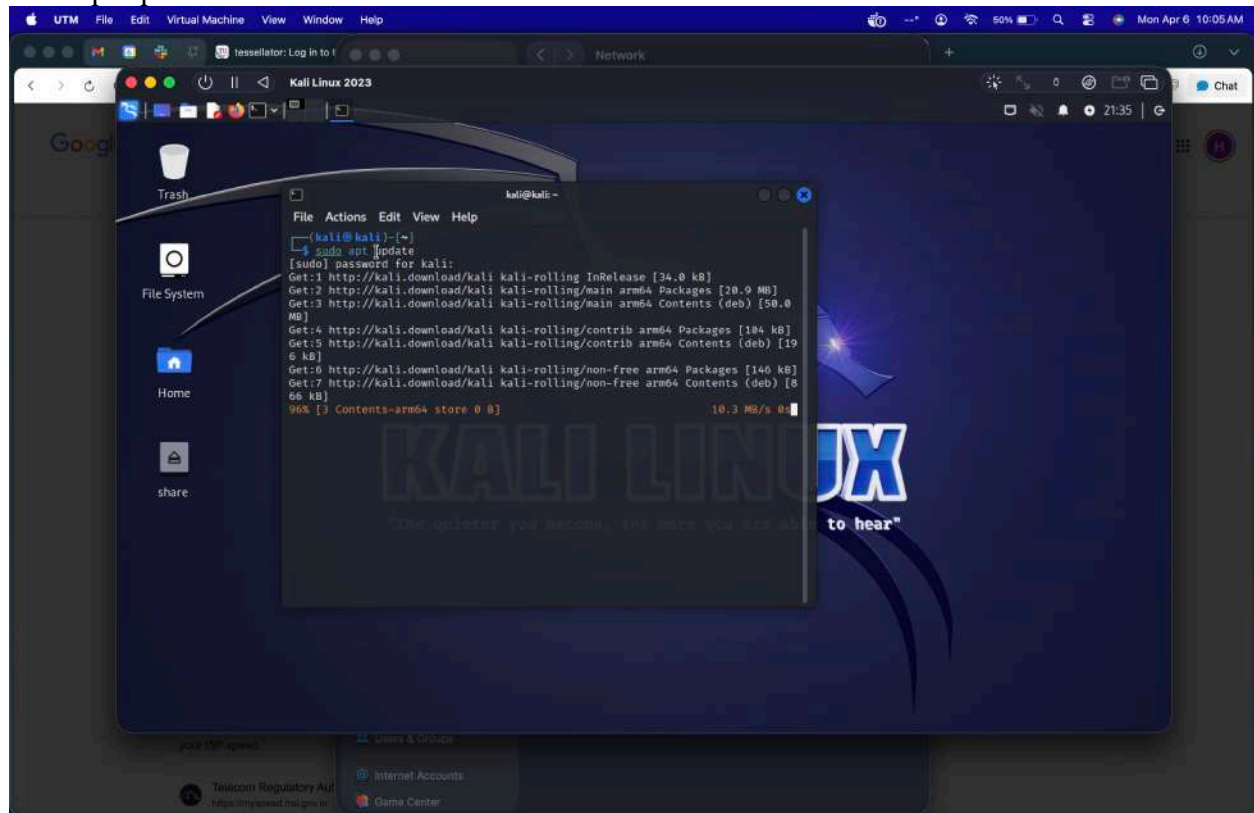
To analyze privacy, network behavior, and security risks of applications and websites using Kali Linux tools.

Tools Required

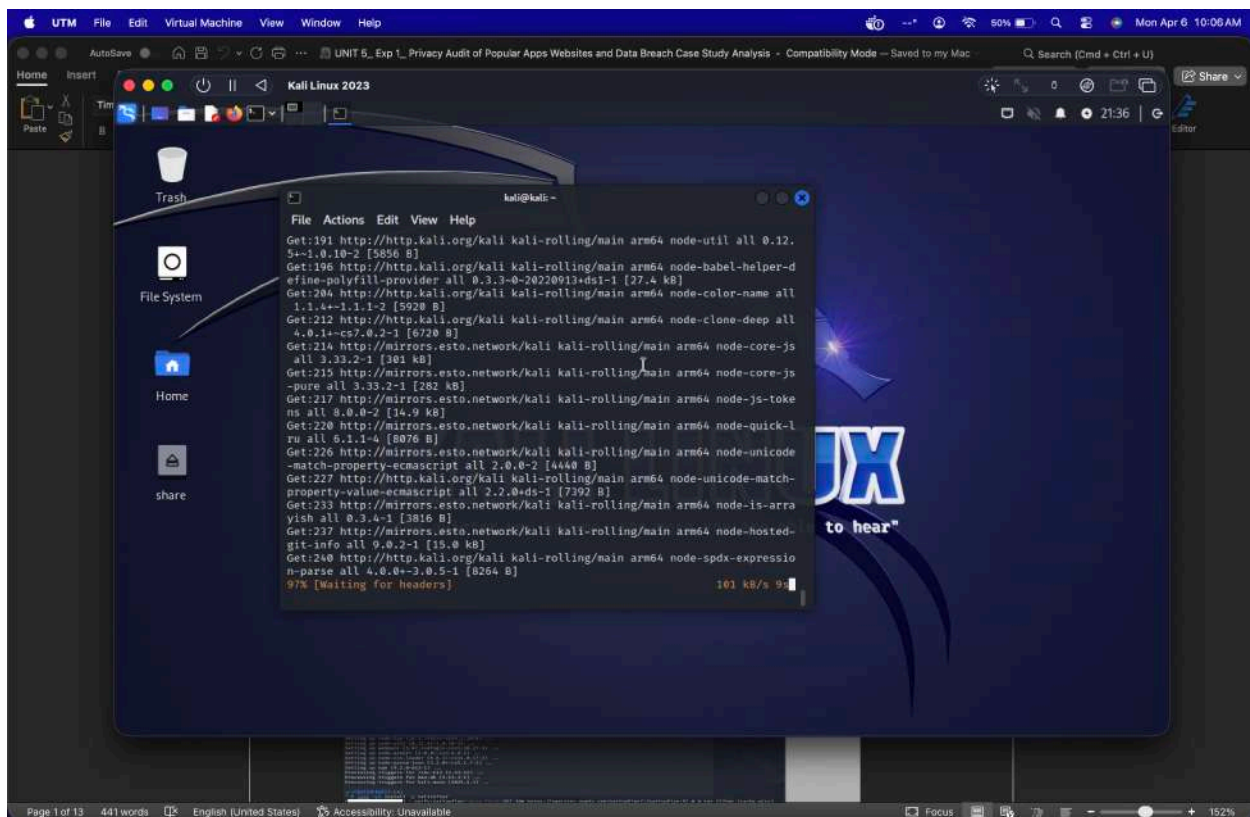
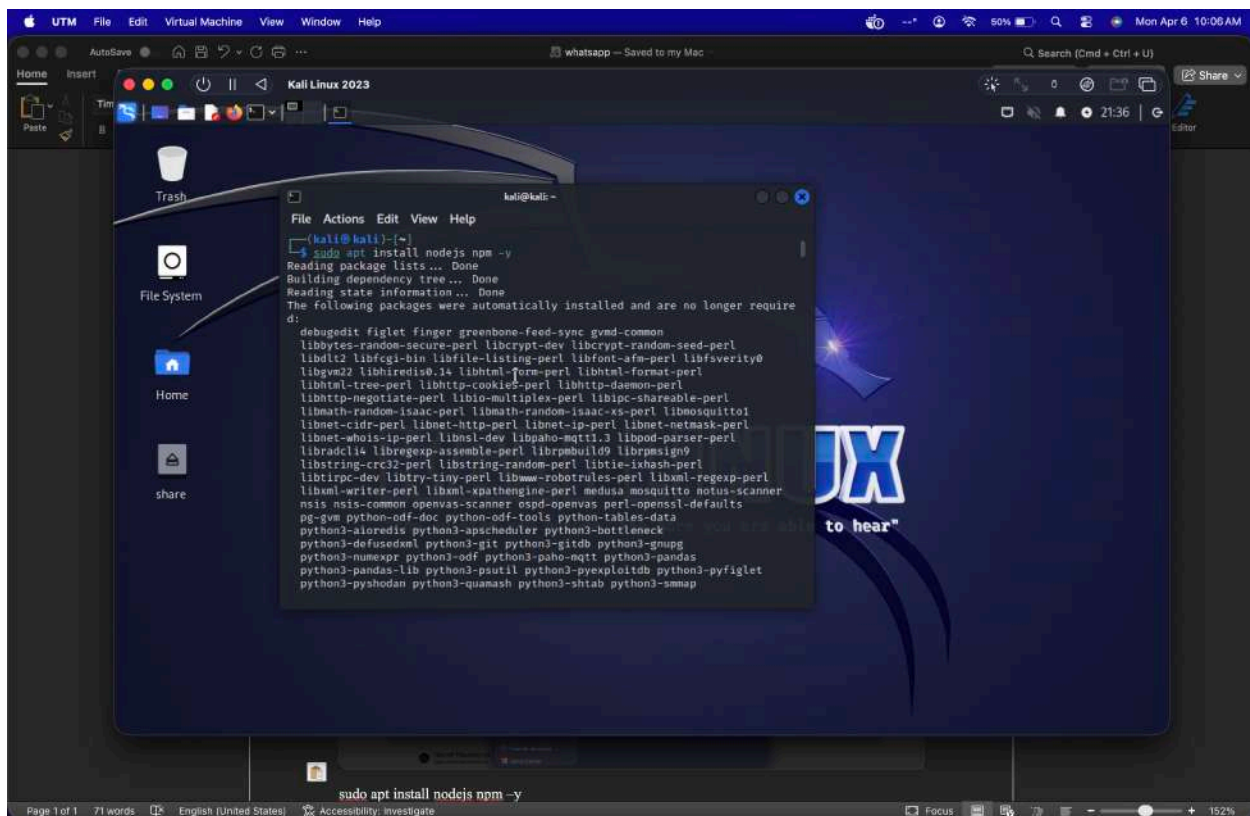
- **Wireshark**
- **Burp Suite**
- Kali Linux (VirtualBox)
- Desktop WhatsApp (Nativefier)
- Web Browser (Firefox)

Step 1: Install Node.js & npm

sudo apt update

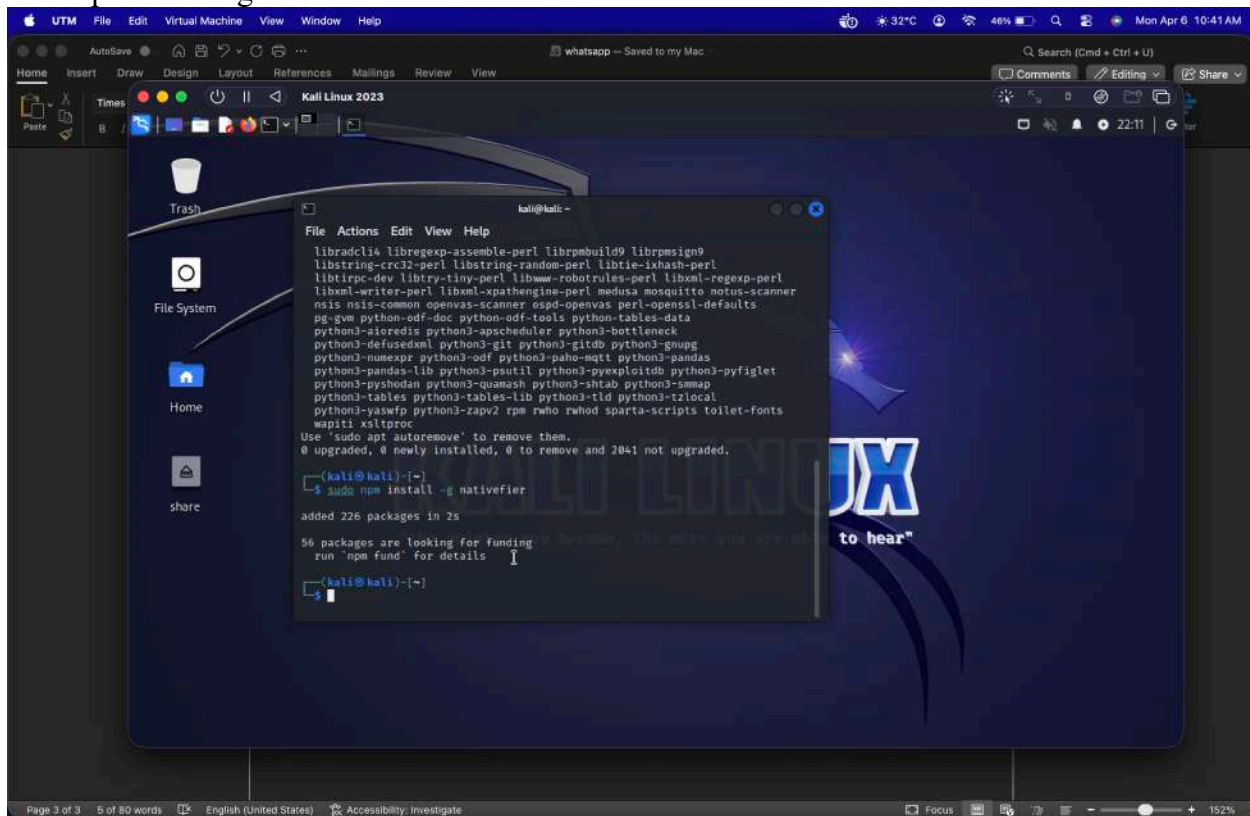


sudo apt install nodejs npm -y

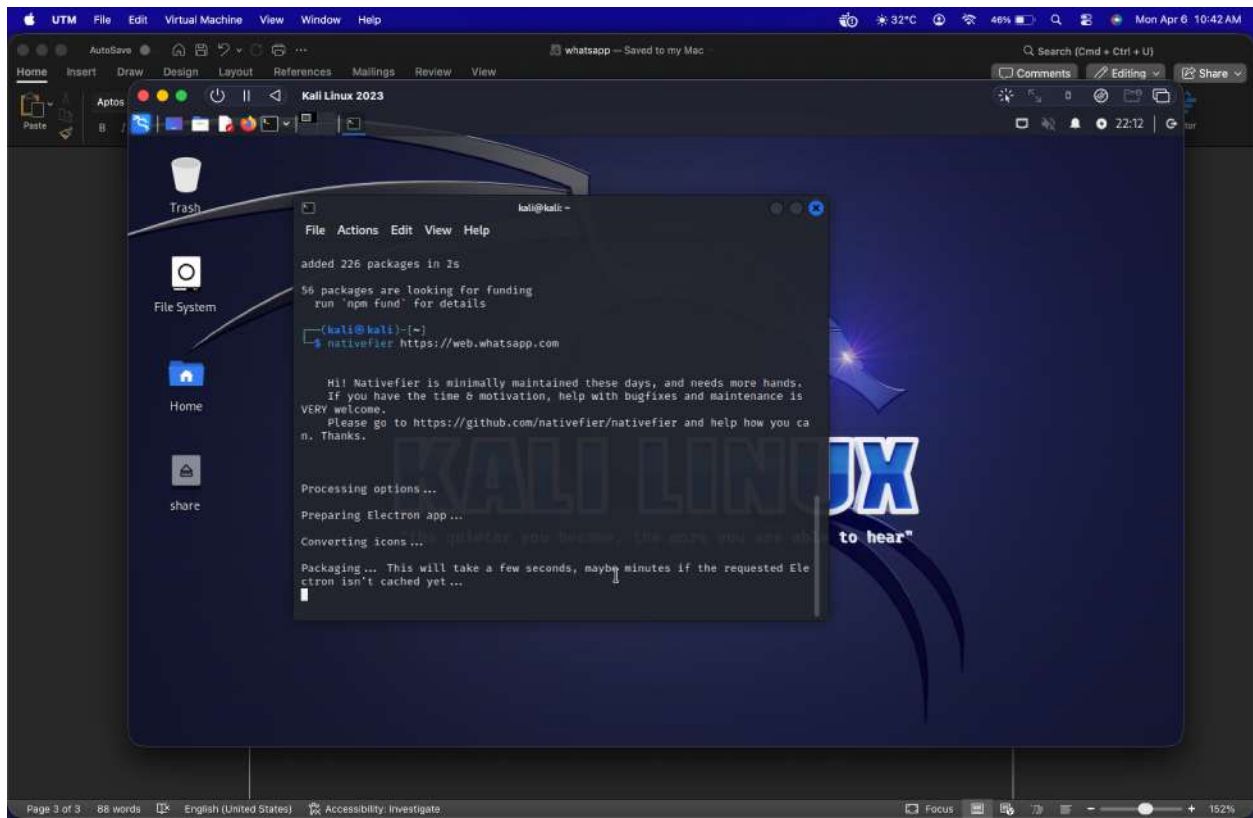


Step 2: Install Nativefier

sudo npm install -g nativefier



Step 3: Create WhatsApp Desktop App
nativefier <https://web.whatsapp.com>

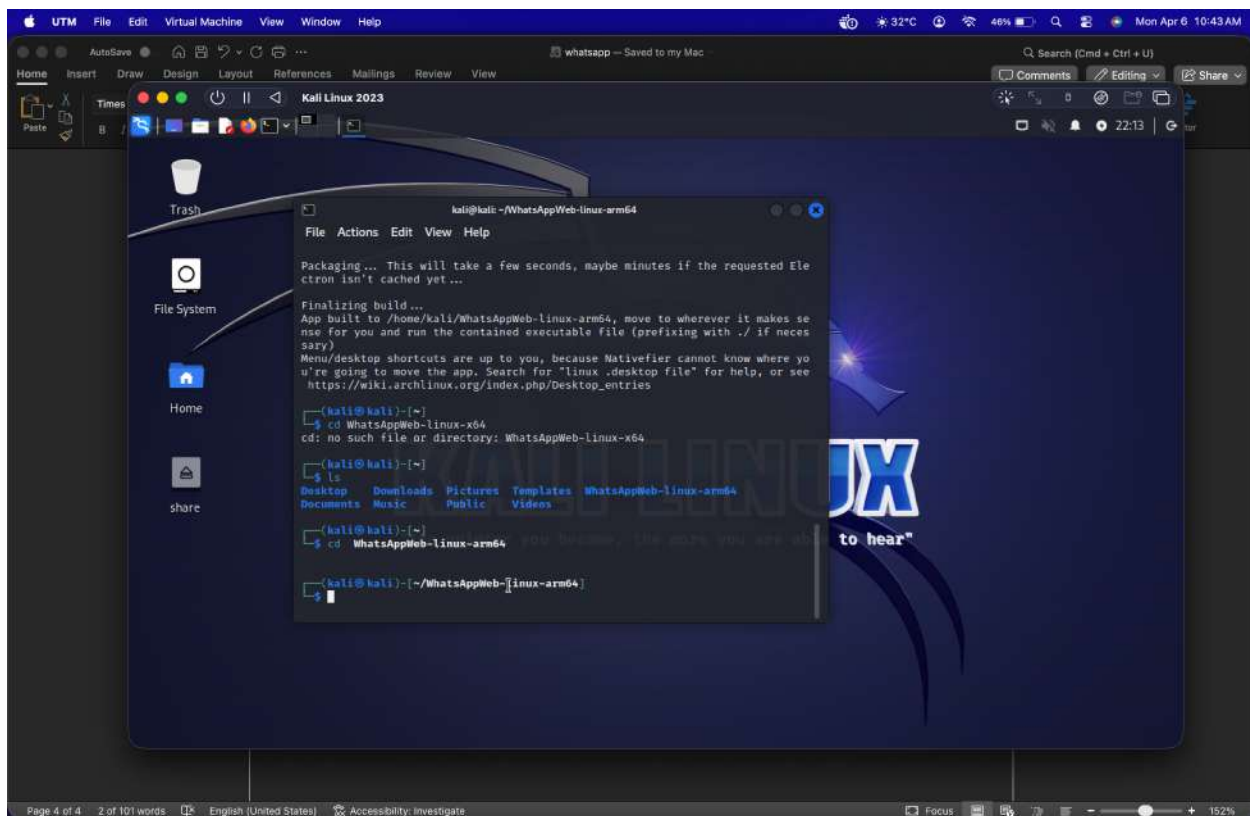


It will create a folder like:

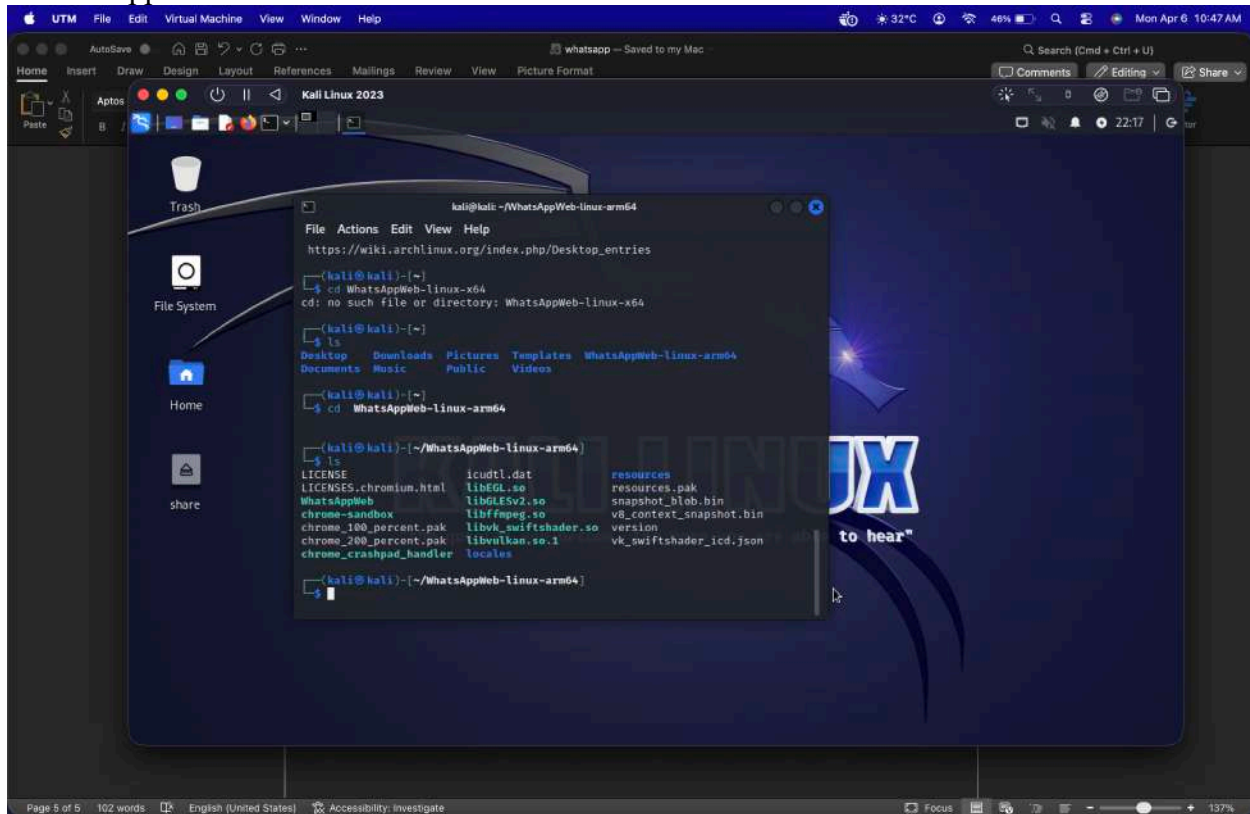
web-whatsapp-linux-x64

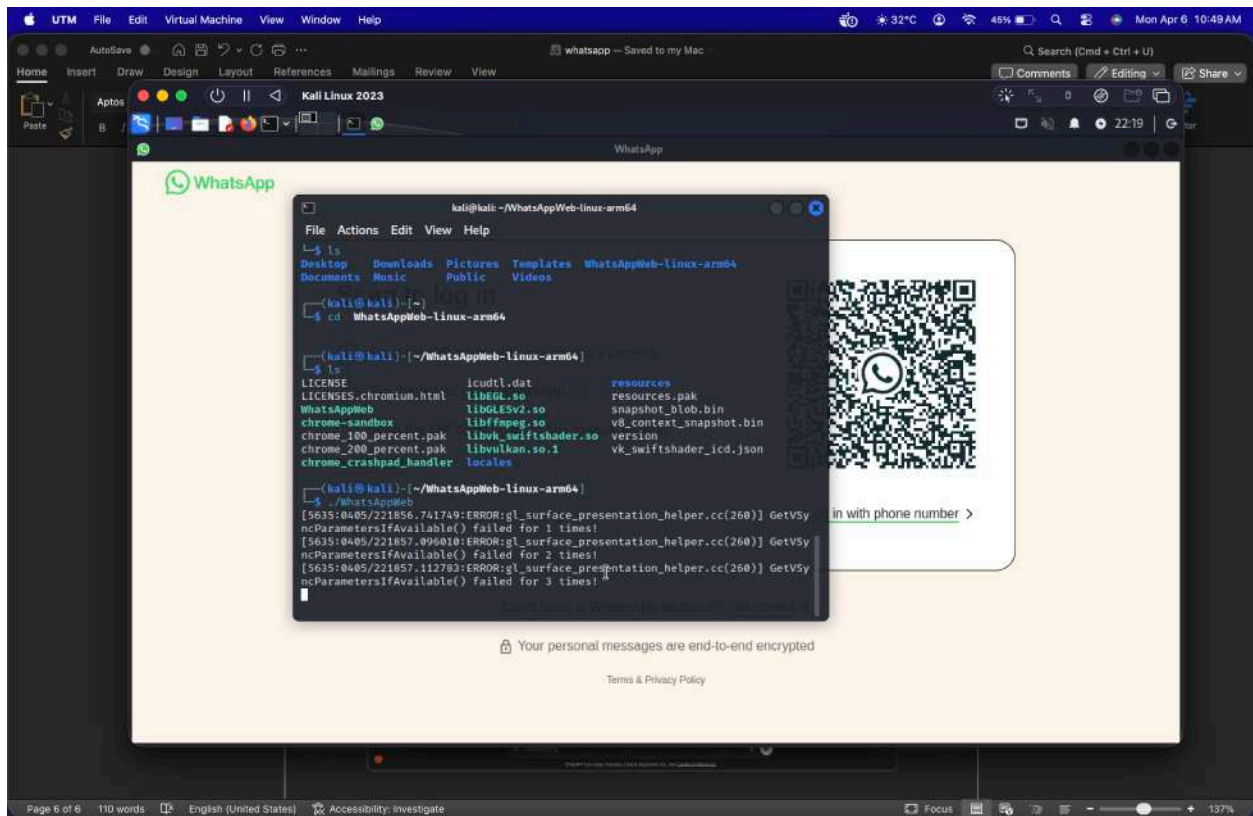
Step 4: Run WhatsApp

cd WhatsAppWeb-linux-x64



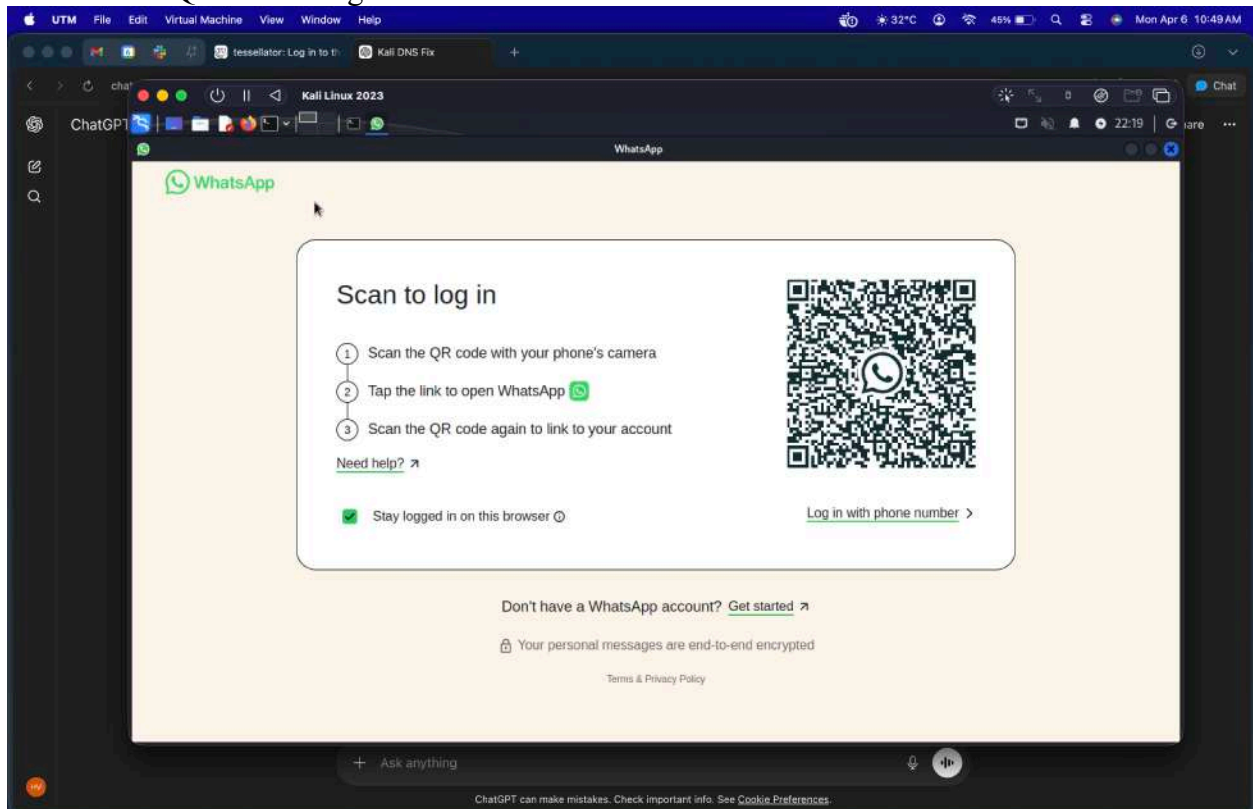
./WhatsAppWeb





Step 5: Login

- Scan QR code using mobile



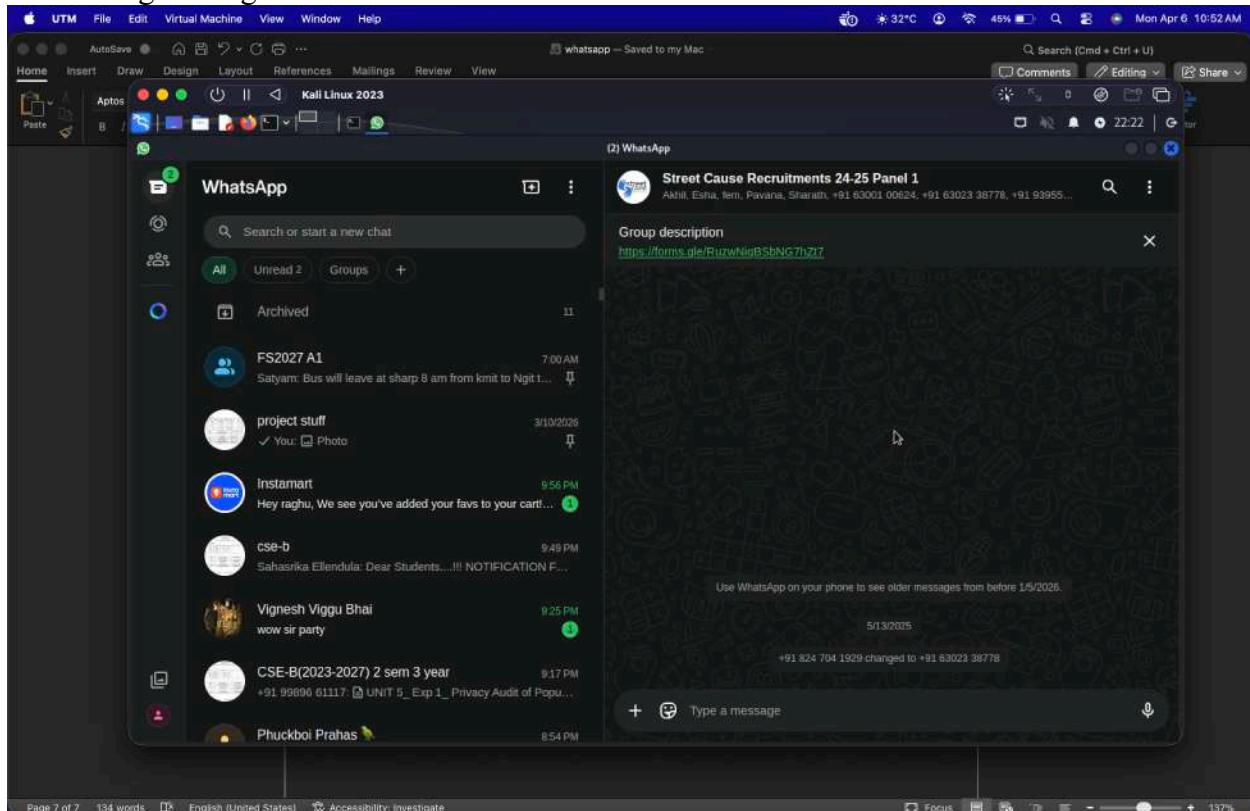
PART 1: Privacy Audit of Desktop WhatsApp

Step 1: Launch WhatsApp Desktop

cd WhatsAppWeb-linux-x64

./WhatsAppWeb

- QR Code appears
- Login using mobile → Linked Devices

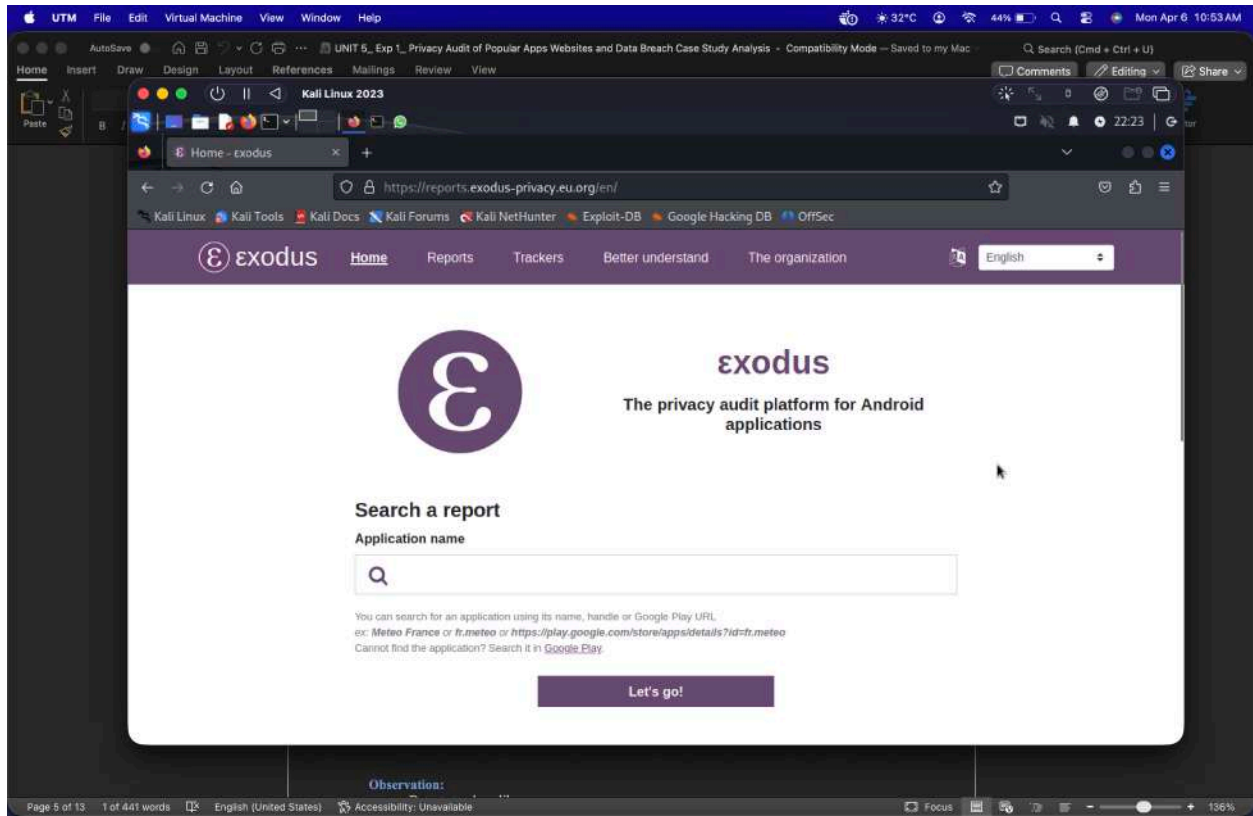


Step 1: Install WhatsApp APK (Optional in Emulator)

- Download APK from trusted source or use your mobile

Step 2: Analyze Trackers using Exodus Privacy

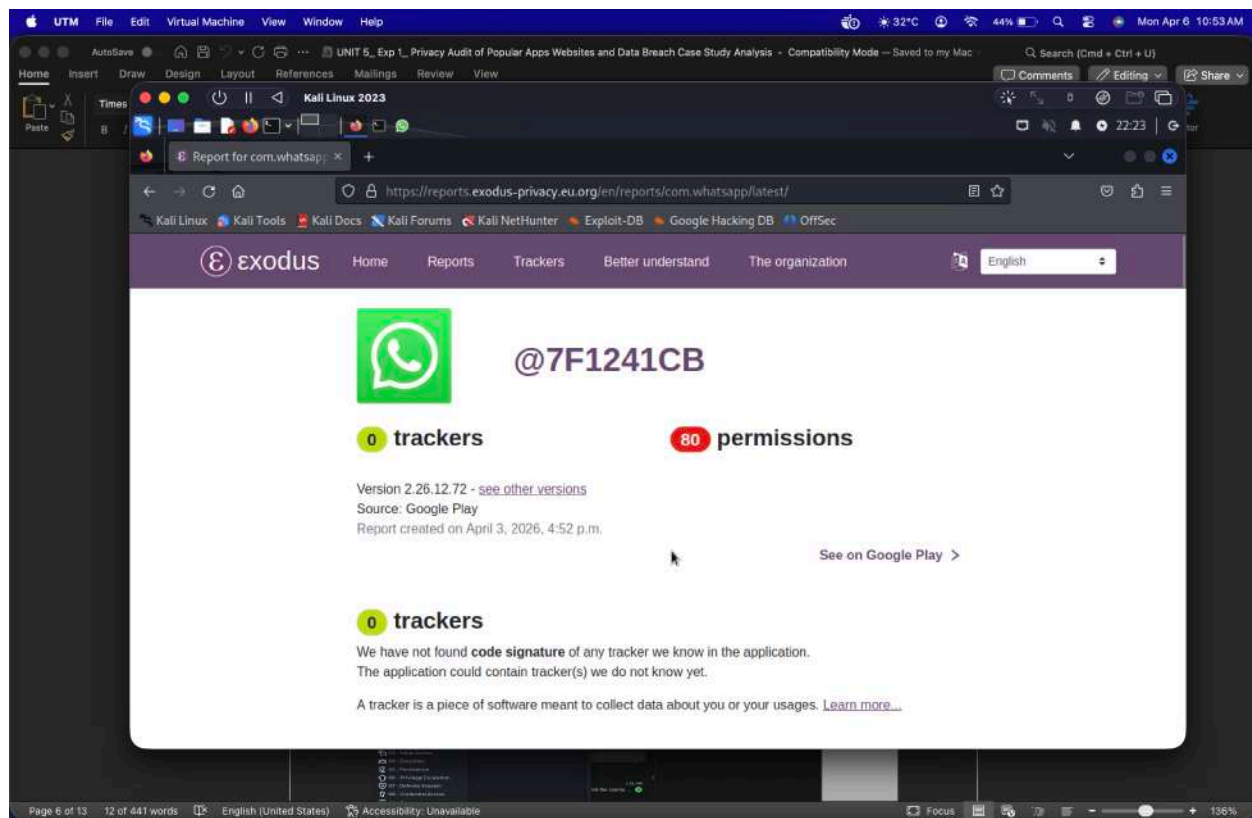
1. Open browser in Kali
2. Go to: <https://reports.exodus-privacy.eu.org>



1. Search for WhatsApp

Observation:

- Detects trackers like:
 - Google Analytics
 - Crashlytics
 - Facebook Analytics

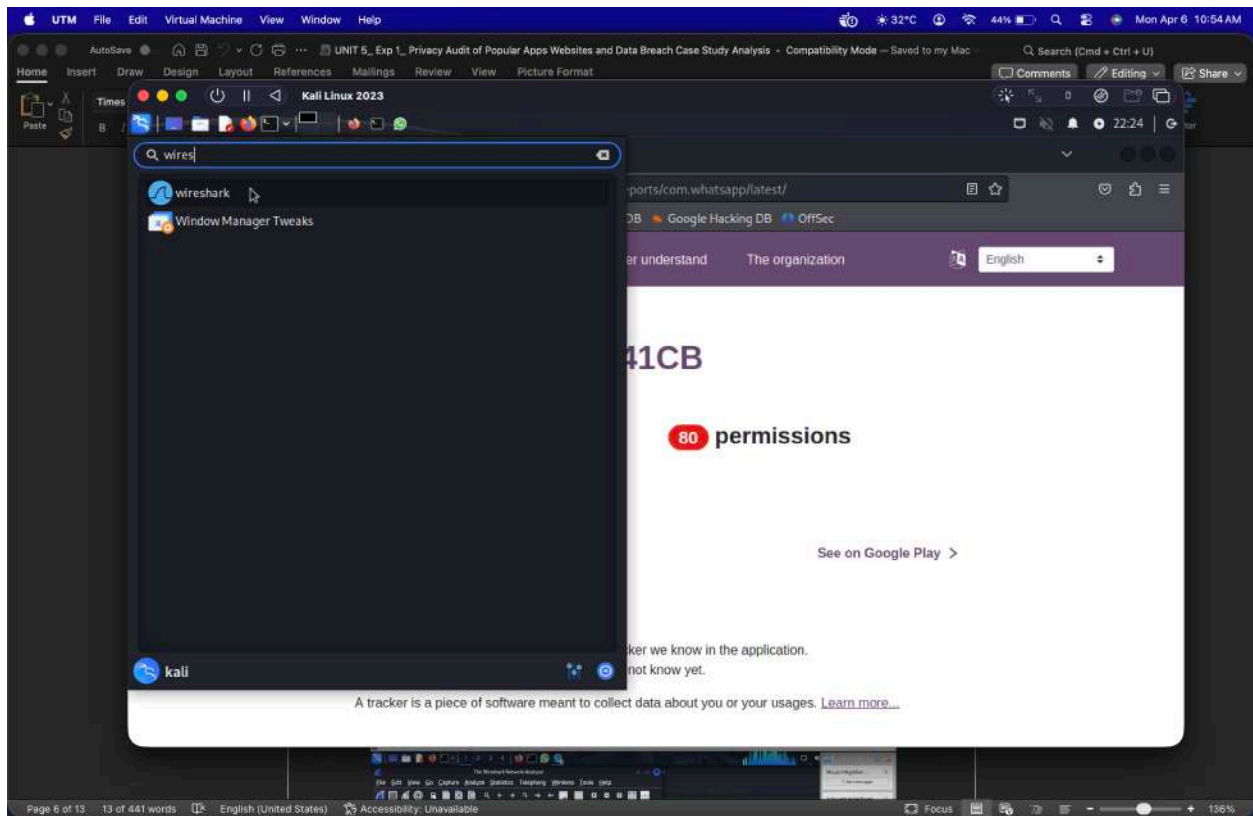


Record in lab:

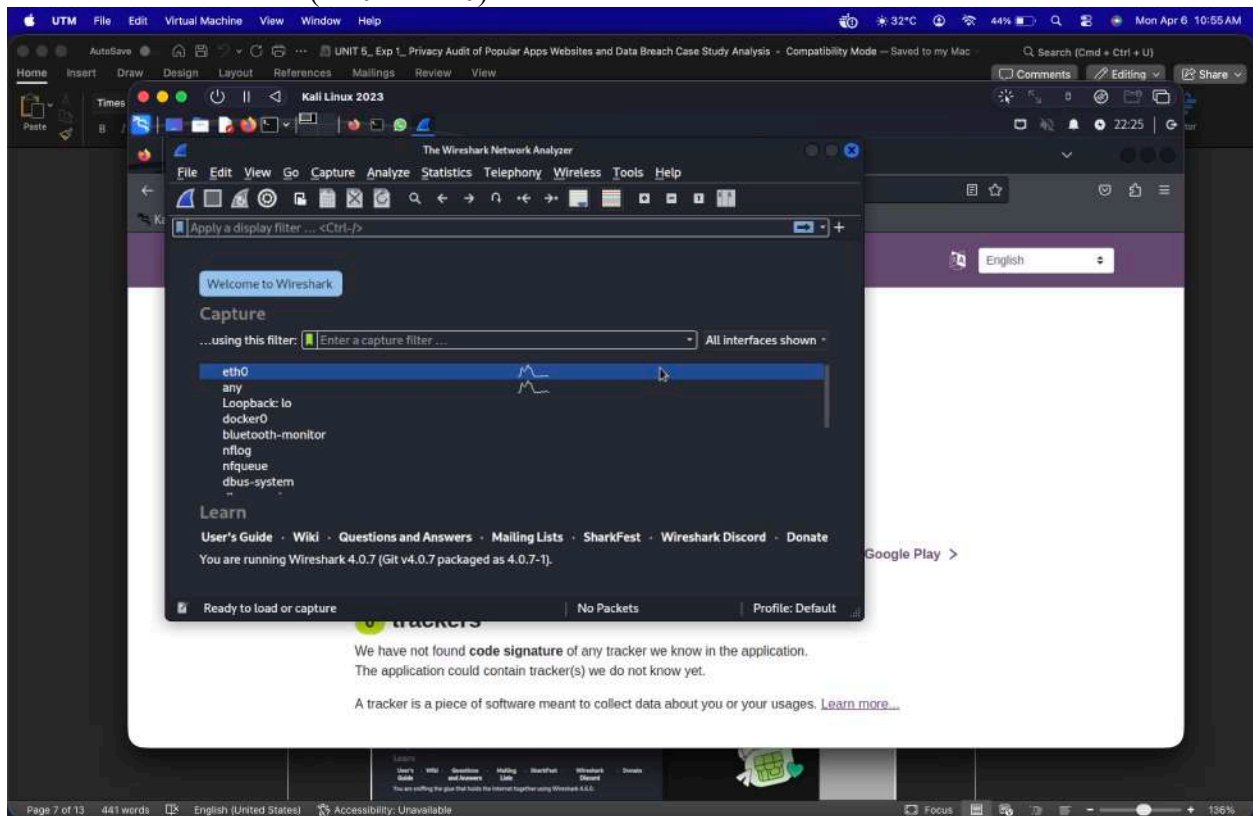
- Number of trackers
- Permissions used

Step 2: Start Wireshark

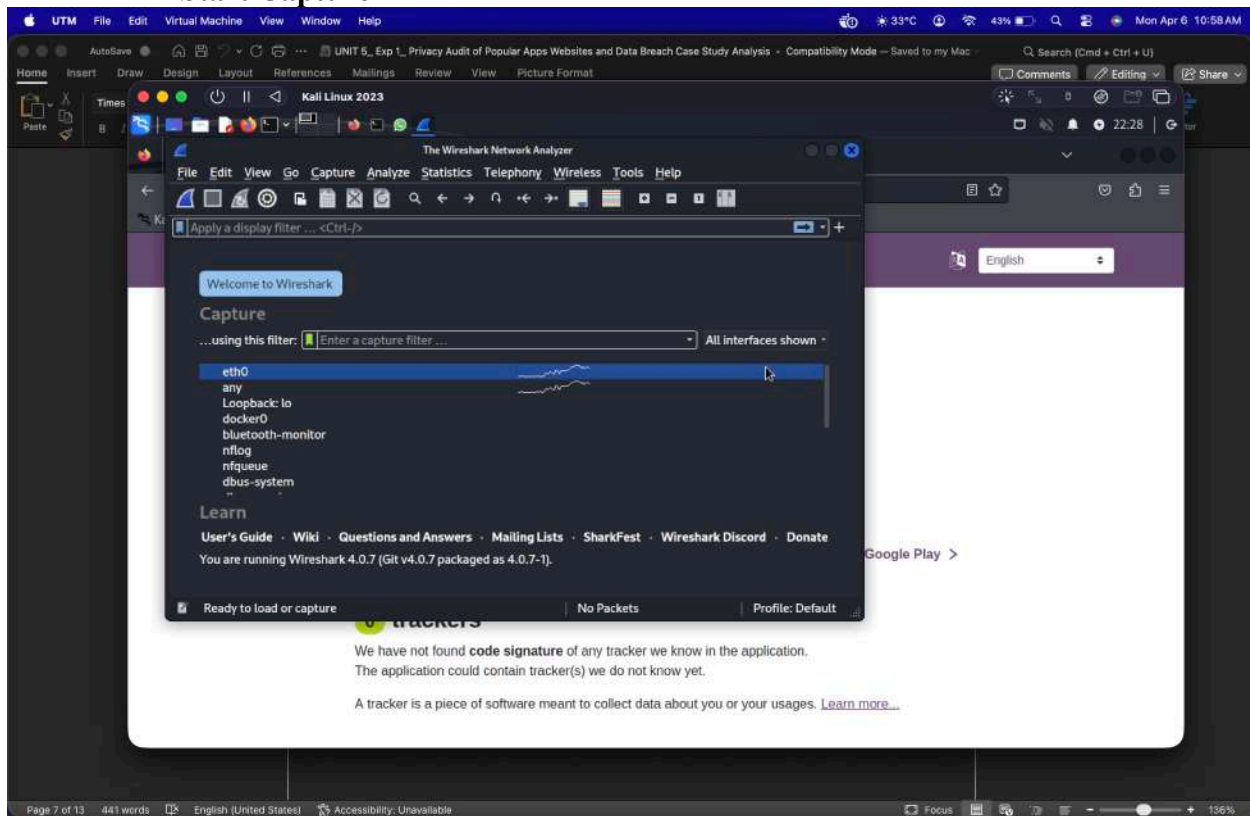
Wireshark



- Select interface (eth0 / wlan0)



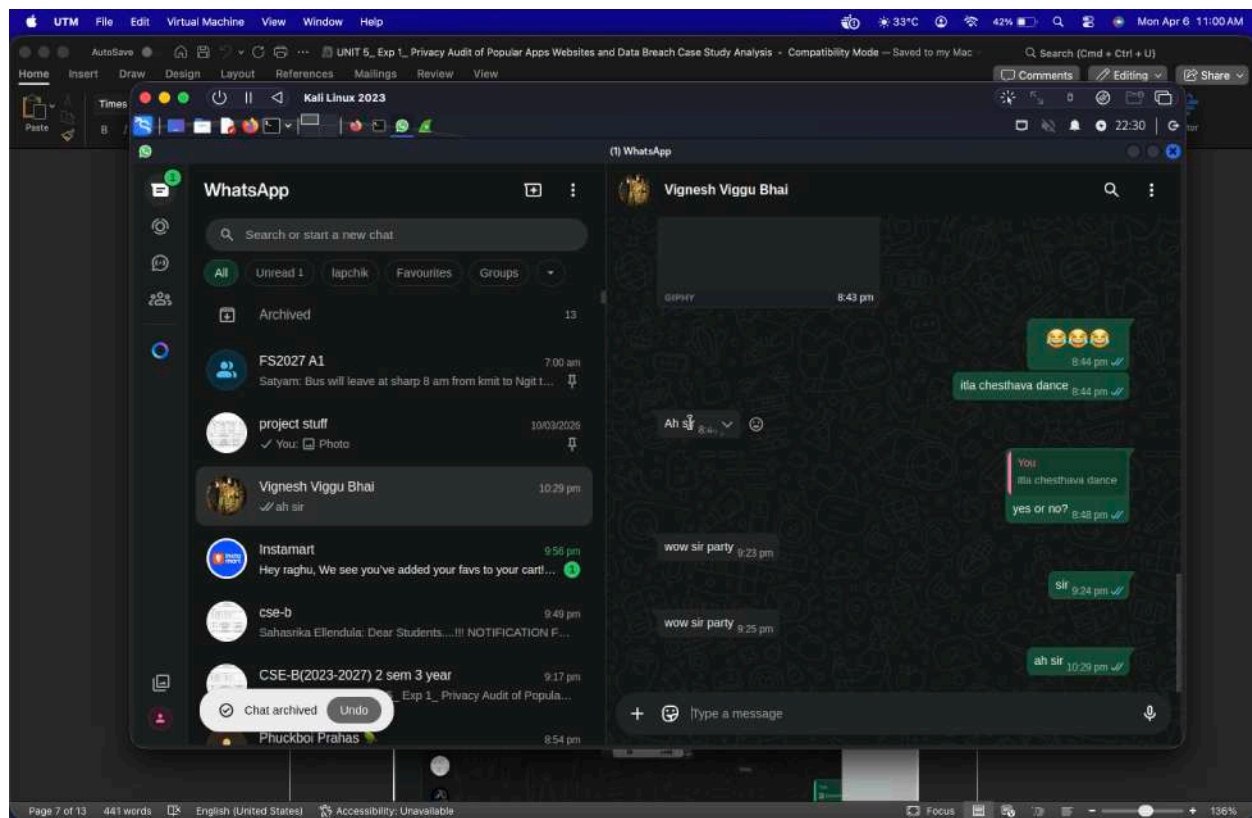
- Click Start Capture



Step 3: Generate Traffic

In WhatsApp Desktop:

- Send messages
- Send images
- Open chats



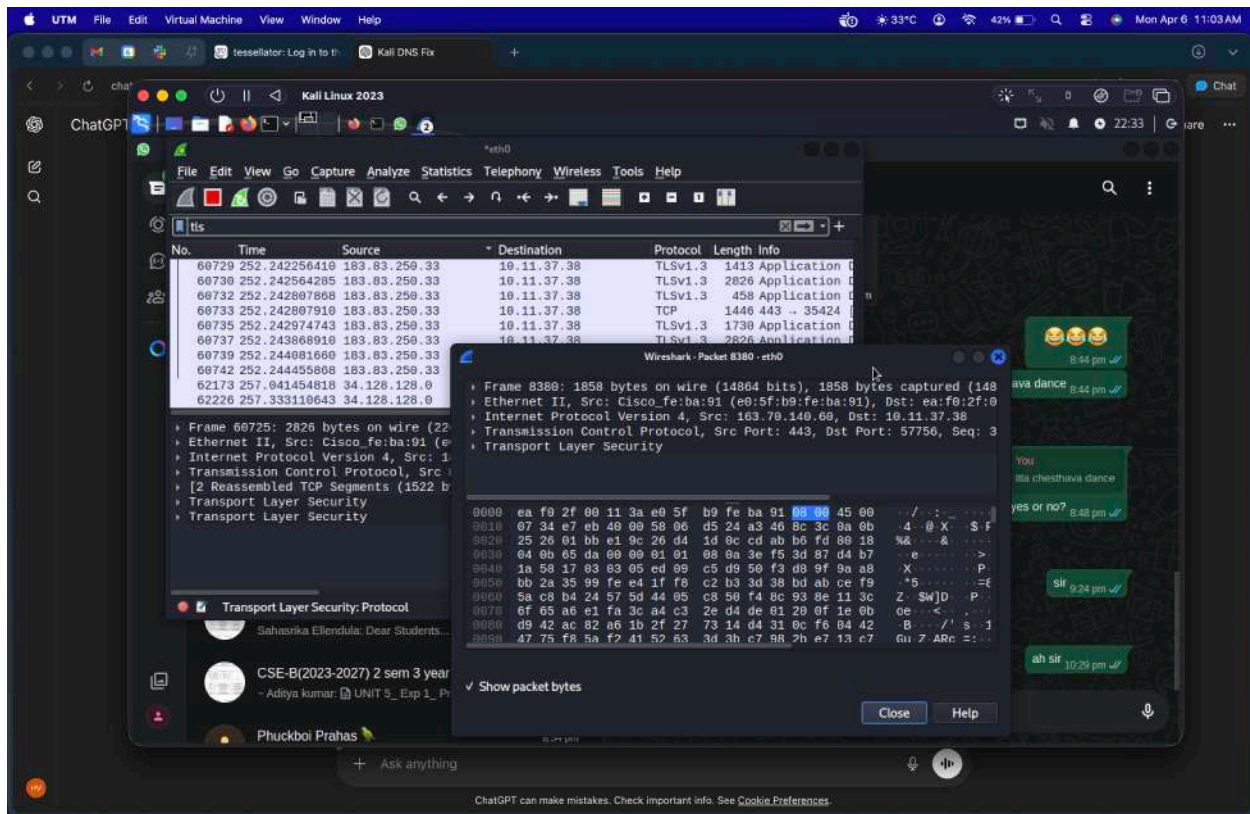
✅ Step 4: Apply Filters

Use filters:

tls

dns

Message content NOT visible



Findings

End-to-End Encryption

Secure communication

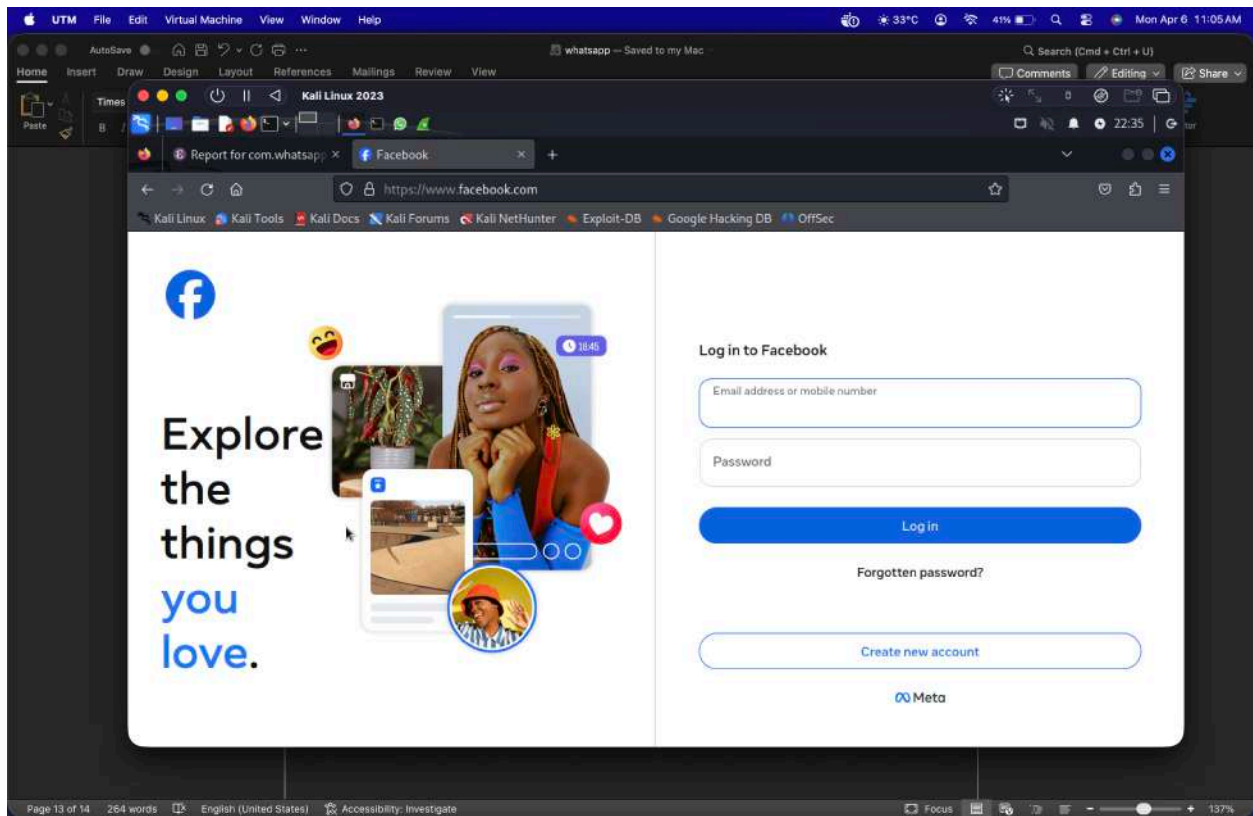
Metadata visible:

- IP address
- Time
- Server communication

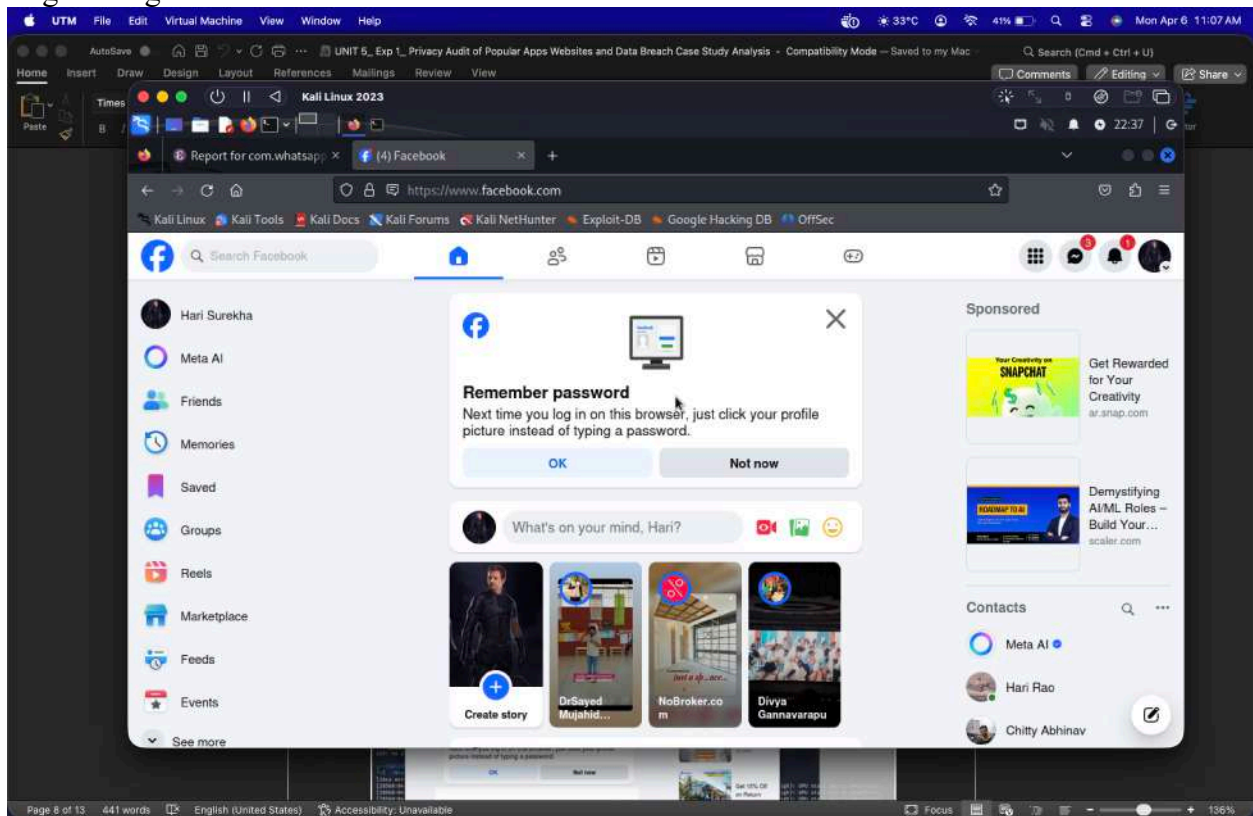
PART 2: Privacy Audit of Facebook Website

Step 1: Open Facebook

<https://www.facebook.com>

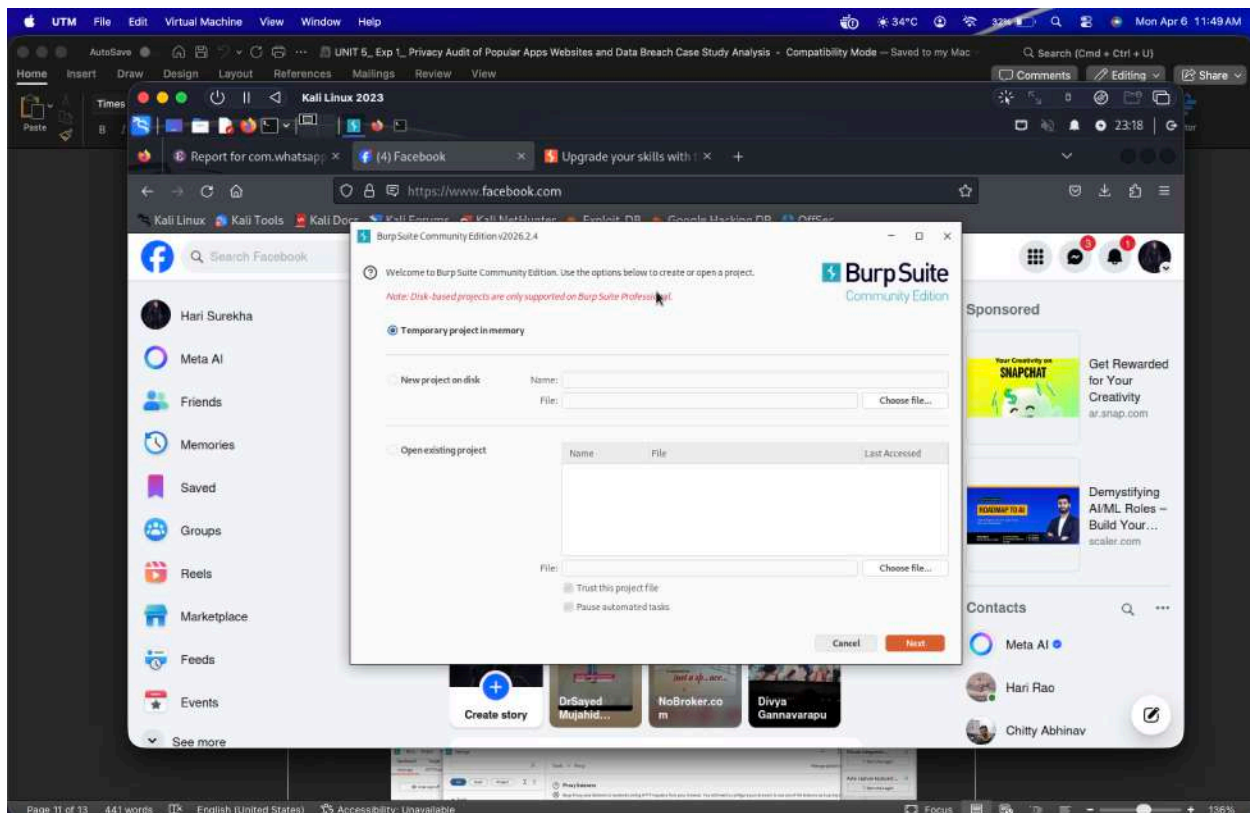
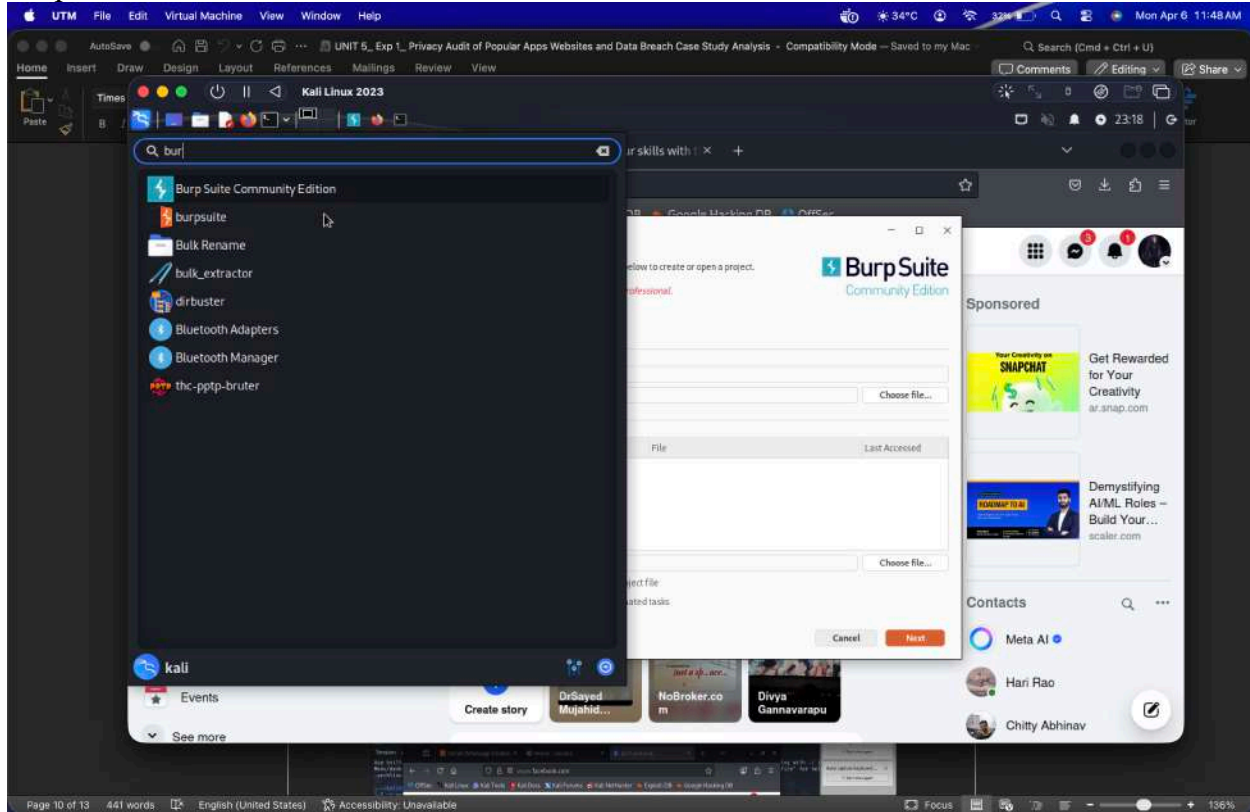


Login using test account



Step 2: Start Burp Suite

Burpsuite

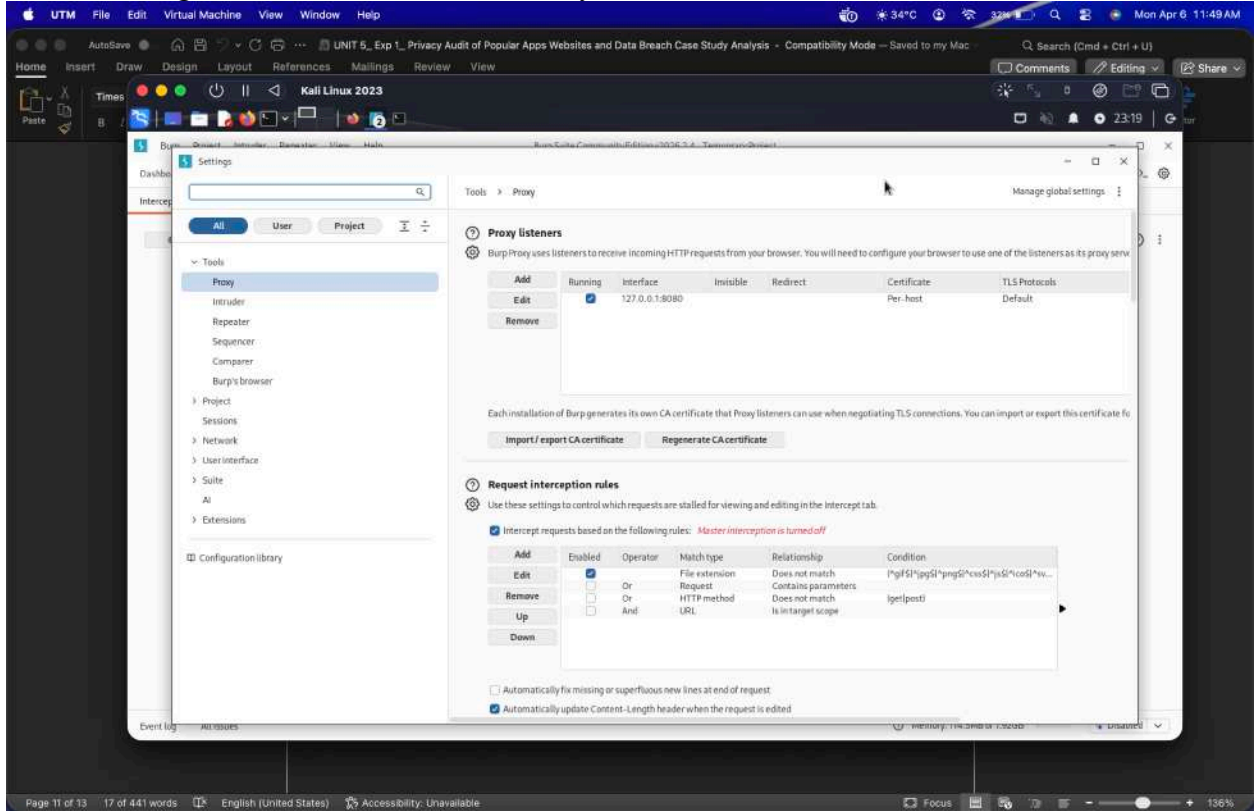


✅ Step 3: Configure Proxy

- IP: 127.0.0.1
- Port: 8080

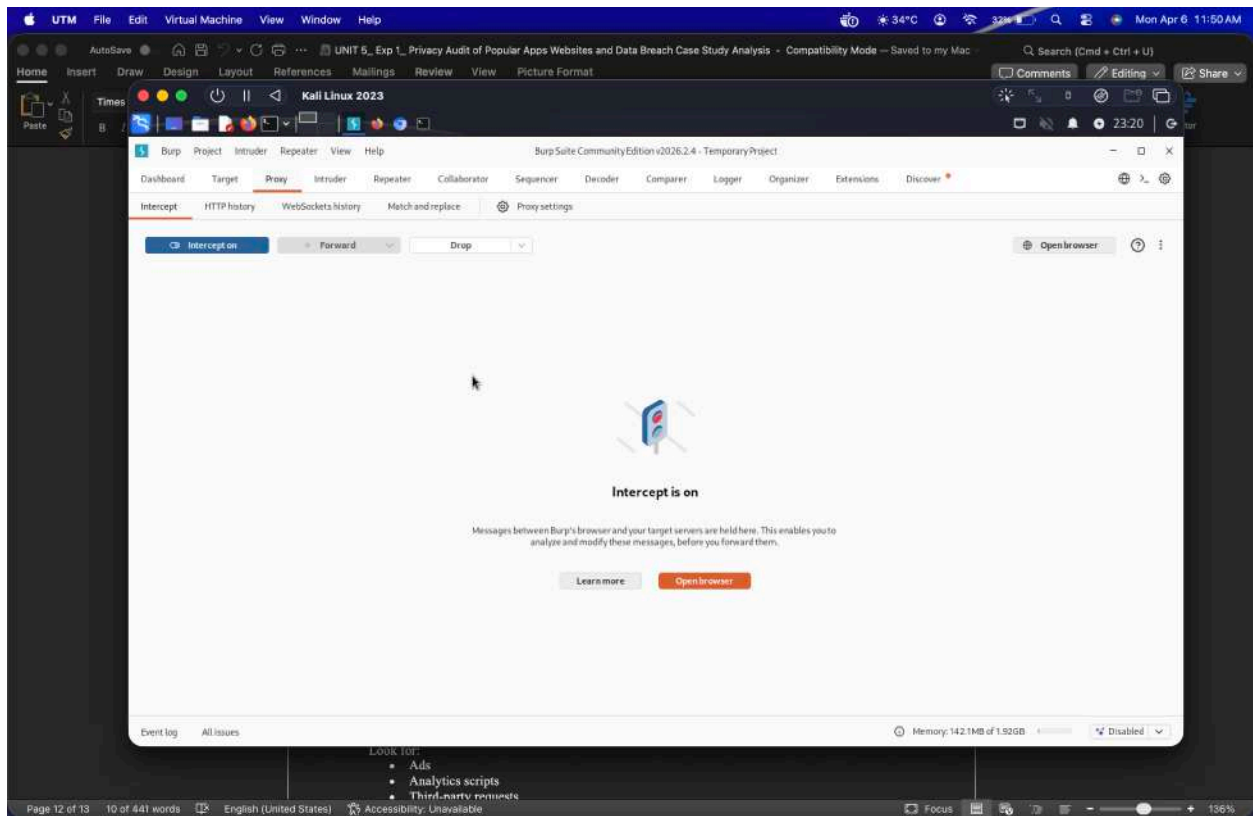
In browser:

- Settings → Network → Manual Proxy



✔ Step 4: Intercept Traffic

- Turn **Intercept ON**
- Reload Facebook



Observe:

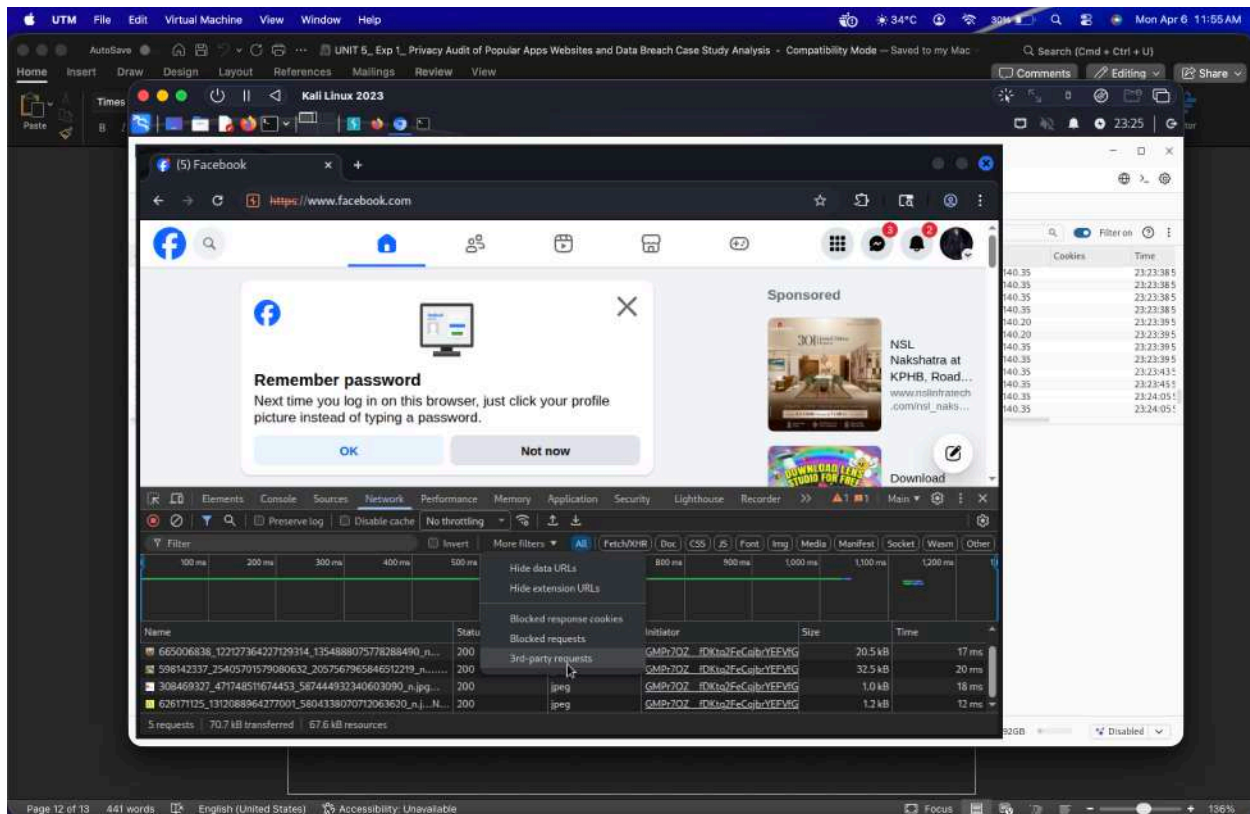
- Cookies (datr, fr)
- Session IDs
- Headers

Step 5: Analyze Tracking

Press **F12** → **Network** tab

Look for:

- Ads
- Analytics scripts
- Third-party requests



Findings

Heavy tracking

Uses cookies for personalization

Third-party integrations

PART 3: Data Breach Case Study

Case Study 1: Facebook Data Breach

- 533 million users affected
- Data leaked:
 - Phone numbers
 - Emails

Cause:

- API vulnerability

Case Study 2: WhatsApp Pegasus Attack

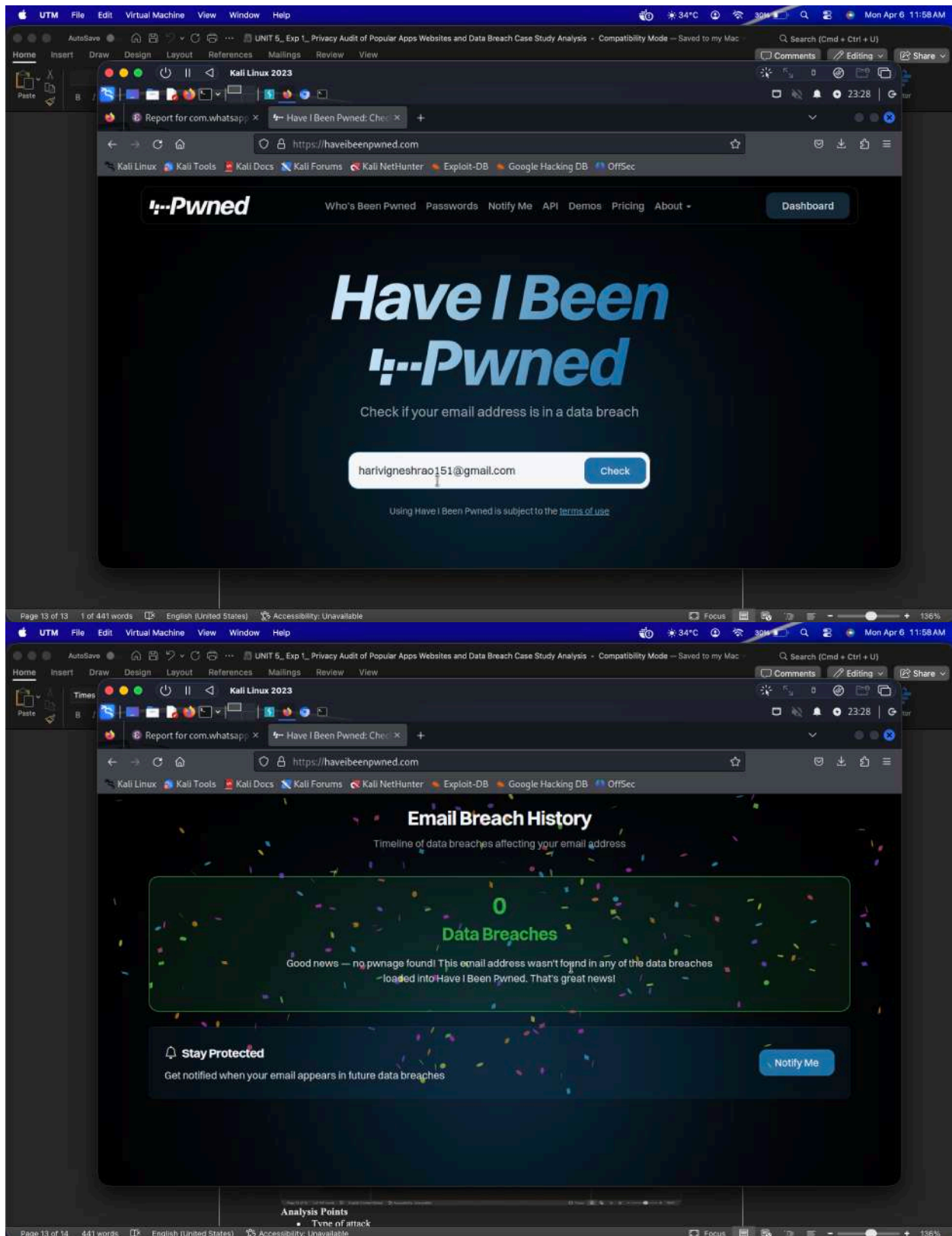
- Spyware attack via missed call
- Remote device access

Step 3: Check Email Breach

Go to:

👉 <https://haveibeenpwned.com>

Enter email → check breach status



Analysis Points

- Type of attack

- Vulnerability
- Data exposed
- Prevention

Observations

- WhatsApp uses strong encryption
- Facebook performs extensive tracking
- Data breaches occur due to vulnerabilities

Result

Privacy audit was successfully performed using Kali Linux tools. WhatsApp ensures secure communication through encryption, while Facebook shows extensive tracking behavior. Data breach analysis highlighted risks of poor security practices.